In [1]:
```python
import pandas as pd
import numpy as np
import seaborn as sea
import matplotlib.pyplot as mtn
```

In [2]:
```python
df = pd.read_csv(r'C:\Users\chira\Desktop\Python CSVs\Sales Data.csv', encoding = 'unicode_escape')
```

In [3]:
```python
df.head(3)
```

Out[3]:

	Order_ID	Customer_ID	Product_Category	Product_Name	Quantity	Unit_Price	Total_Price	Purchase_Date	Payment_Method	Shipping_Time_Days	Customer_Rating	Discount_Percentage
0	ORD100000	CUST6193	Books	Sofa	3	97.51	292.52	2022-01-01 00:00:00	Debit Card	8	3	19
1	ORD100001	CUST4745	Books	Sofa	4	265.08	1060.31	2022-01-01 01:00:00	Bitcoin	9	4	13
2	ORD100002	CUST2838	Toys	Novel	1	82.31	82.31	2022-01-01 02:00:00	Credit Card	8	5	29

In [4]:
```python
pd.isnull(df).sum()
```

Out[4]:
```
Order_ID               0
Customer_ID            0
Product_Category       0
Product_Name           0
Quantity               0
Unit_Price             0
Total_Price            0
Purchase_Date          0
Payment_Method         0
Shipping_Time_Days     0
Customer_Rating        0
Discount_Percentage    0
dtype: int64
```

In [5]:
```python
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7000 entries, 0 to 6999
Data columns (total 12 columns):
 #   Column               Non-Null Count  Dtype  
---  ------               --------------  -----  
 0   Order_ID             7000 non-null   object 
 1   Customer_ID          7000 non-null   object 
 2   Product_Category     7000 non-null   object 
 3   Product_Name         7000 non-null   object 
 4   Quantity             7000 non-null   int64  
 5   Unit_Price           7000 non-null   float64
 6   Total_Price          7000 non-null   float64
 7   Purchase_Date        7000 non-null   object 
 8   Payment_Method       7000 non-null   object 
 9   Shipping_Time_Days   7000 non-null   int64  
 10  Customer_Rating      7000 non-null   int64  
 11  Discount_Percentage  7000 non-null   int64  
dtypes: float64(2), int64(4), object(6)
memory usage: 656.4+ KB
```

In [6]:
```python
df['Purchase_Date'] = pd.to_datetime(df["Purchase_Date"], format = 'mixed')
```

In [7]:
```python
df["Order_Date"] = df["Purchase_Date"].dt.date
```

1. What is the overall revenue trend over time?

In [8]:
```python
df.head(2)
```

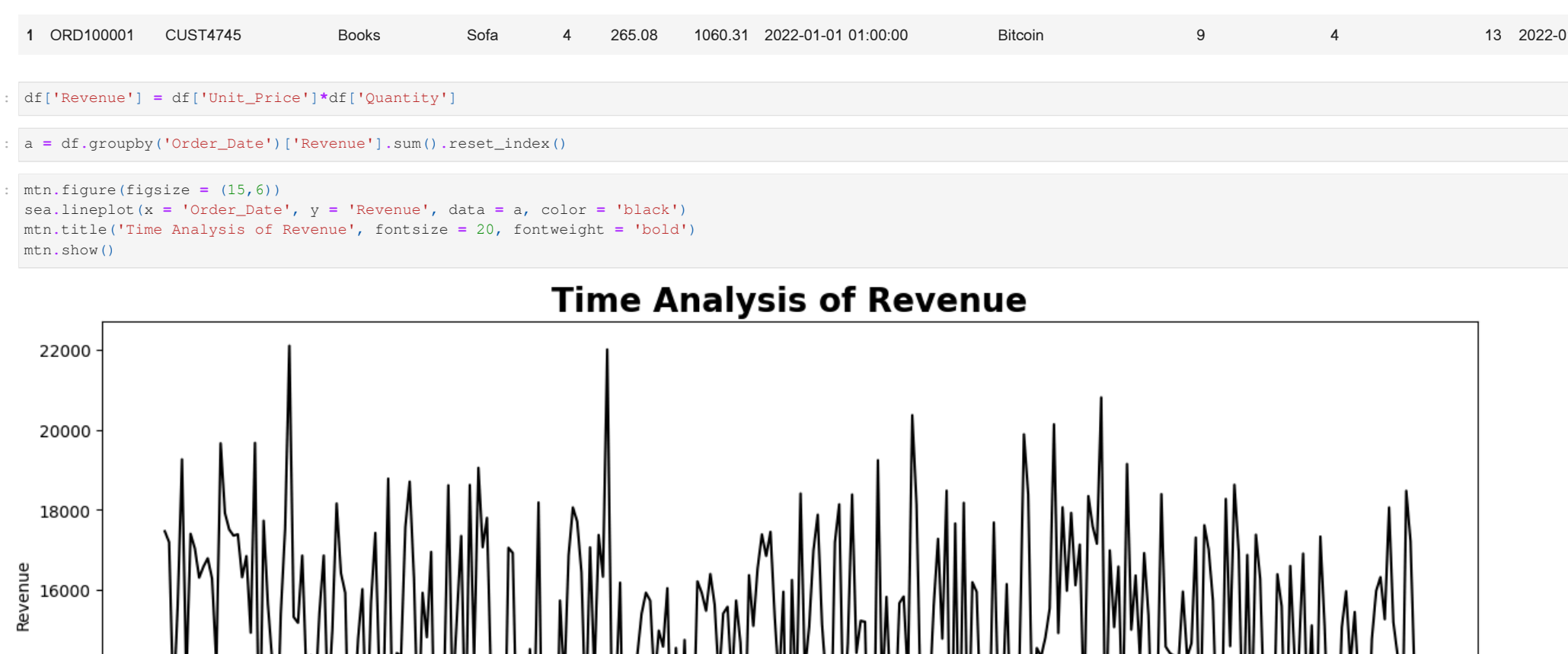
Out[8]:

	Order_ID	Customer_ID	Product_Category	Product_Name	Quantity	Unit_Price	Total_Price	Purchase_Date	Payment_Method	Shipping_Time_Days	Customer_Rating	Discount_Percentage	Order_Date
0	ORD100000	CUST6193	Books	Sofa	3	97.51	292.52	2022-01-01 00:00:00	Debit Card	8	3	19	2022-01-01
1	ORD100001	CUST4745	Books	Sofa	4	265.08	1060.31	2022-01-01 01:00:00	Bitcoin	9	4	13	2022-01-01

In [9]:
```python
df['Revenue'] = df['Unit_Price']*df['Quantity']
```

In [10]:
```python
a = df.groupby('Order_Date')['Revenue'].sum().reset_index()
```

In [11]:
```python
mtn.figure(figsize = (15,6))
sea.lineplot(x = 'Order_Date', y = 'Revenue', data = a, color = 'black')
mtn.title('Time Analysis of Revenue', fontsize = 20, fontweight = 'bold')
mtn.show()
```



2. Which product category contributes the most to total sales?

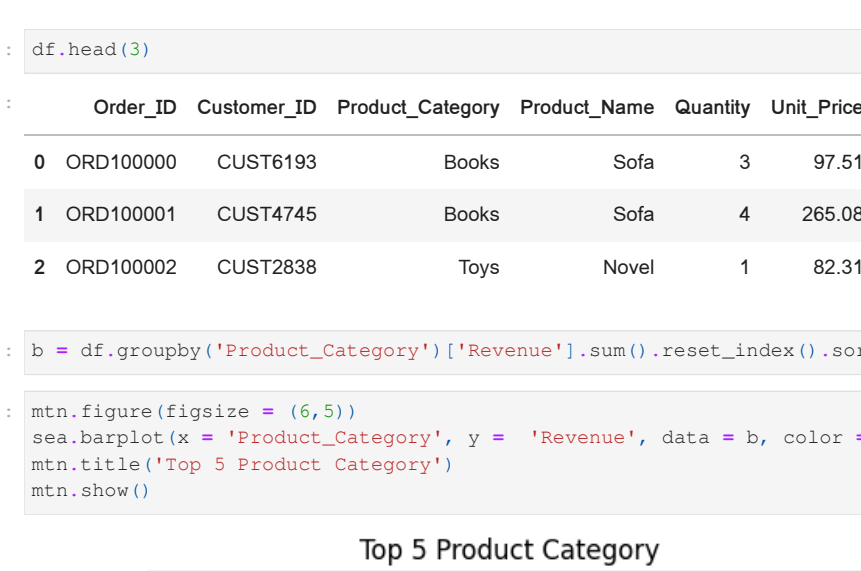
In [12]:
```python
df.head(3)
```

Out[12]:

	Order_ID	Customer_ID	Product_Category	Product_Name	Quantity	Unit_Price	Total_Price	Purchase_Date	Payment_Method	Shipping_Time_Days	Customer_Rating	Discount_Percentage	Order_Date	Revenue
0	ORD100000	CUST6193	Books	Sofa	3	97.51	292.52	2022-01-01 00:00:00	Debit Card	8	3	19	2022-01-01	292.53
1	ORD100001	CUST4745	Books	Sofa	4	265.08	1060.31	2022-01-01 01:00:00	Bitcoin	9	4	13	2022-01-01	1060.32
2	ORD100002	CUST2838	Toys	Novel	1	82.31	82.31	2022-01-01 02:00:00	Credit Card	8	5	29	2022-01-01	82.31

In [13]:
```python
b = df.groupby('Product_Category')['Revenue'].sum().reset_index().sort_values('Revenue', ascending = False).head(5)
```

In [14]:
```python
mtn.figure(figsize = (6,5))
sea.barplot(x = 'Product_Category', y =  'Revenue', data = b, color = 'orange')
mtn.title('Top 5 Product Category')
mtn.show()
```



3. What are the top 5 best-selling products in terms of revenue?

In [15]:
```python
df.head(3)
```

Out[15]:

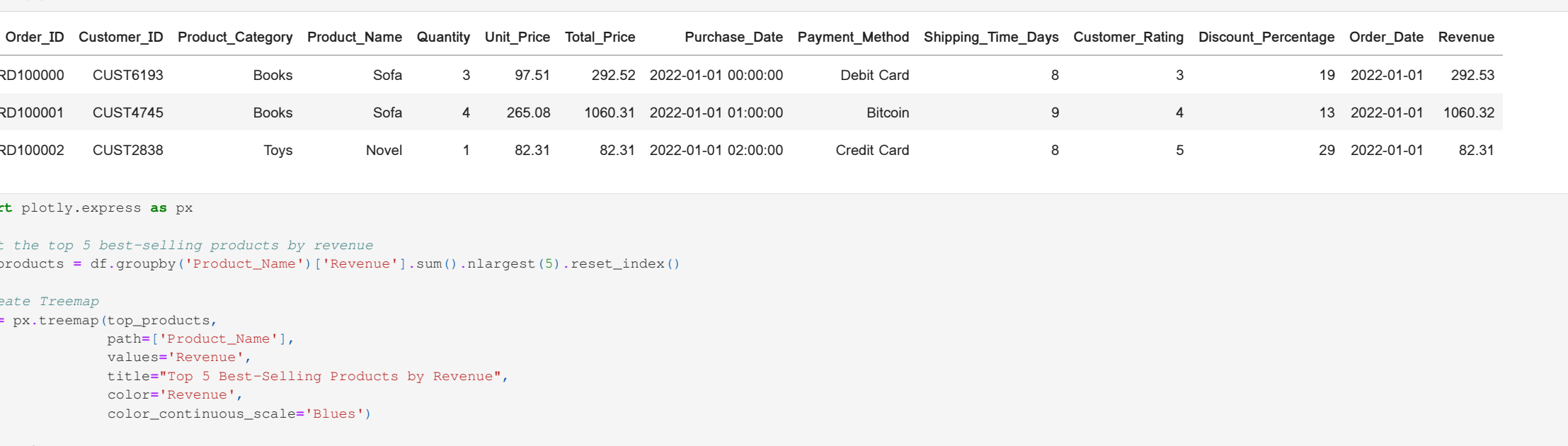
	Order_ID	Customer_ID	Product_Category	Product_Name	Quantity	Unit_Price	Total_Price	Purchase_Date	Payment_Method	Shipping_Time_Days	Customer_Rating	Discount_Percentage	Order_Date	Revenue
0	ORD100000	CUST6193	Books	Sofa	3	97.51	292.52	2022-01-01 00:00:00	Debit Card	8	3	19	2022-01-01	292.53
1	ORD100001	CUST4745	Books	Sofa	4	265.08	1060.31	2022-01-01 01:00:00	Bitcoin	9	4	13	2022-01-01	1060.32
2	ORD100002	CUST2838	Toys	Novel	1	82.31	82.31	2022-01-01 02:00:00	Credit Card	8	5	29	2022-01-01	82.31

In [16]:
```python
import plotly.express as px

# Get the top 5 best-selling products by revenue
top_products = df.groupby('Product_Name')['Revenue'].sum().nlargest(5).reset_index()

# Create Treemap
fig = px.treemap(top_products,
                 path=['Product_Name'],
                 values='Revenue',
                 title="Top 5 Best-Selling Products by Revenue",
                 color='Revenue',
                 color_continuous_scale='Blues')

# Show the Treemap
fig.show()
```



4. How does the average order value (AOV) vary across payment methods?

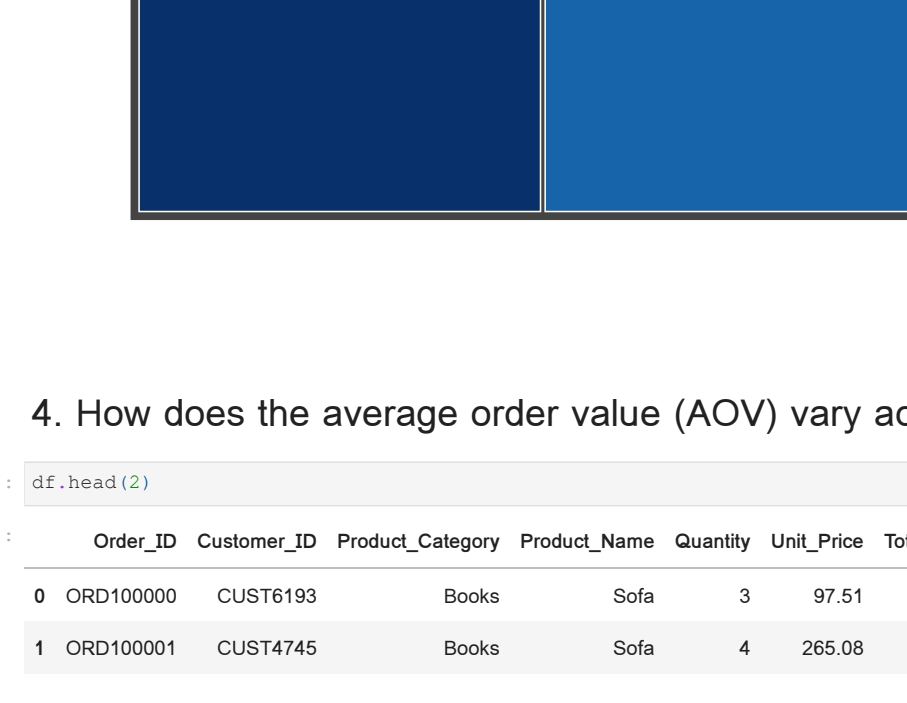
In [17]:
```python
df.head(2)
```

Out[17]:

	Order_ID	Customer_ID	Product_Category	Product_Name	Quantity	Unit_Price	Total_Price	Purchase_Date	Payment_Method	Shipping_Time_Days	Customer_Rating	Discount_Percentage	Order_Date	Revenue
0	ORD100000	CUST6193	Books	Sofa	3	97.51	292.52	2022-01-01 00:00:00	Debit Card	8	3	19	2022-01-01	292.53
1	ORD100001	CUST4745	Books	Sofa	4	265.08	1060.31	2022-01-01 01:00:00	Bitcoin	9	4	13	2022-01-01	1060.32

In [18]:
```python
AOV = round(df.groupby('Payment_Method')['Total_Price'].mean(),1)
```

In [19]:
```python
mtn.figure(figsize = (7,5))
ax = sea.barplot(x = AOV.index, y = AOV.values, color = 'brown')
ax.bar_label(ax.containers[0])
mtn.title("AOV based on Payment Method", fontsize = 16, fontweight = 'bold')
mtn.show()
```



5. What is the most common purchase time for different product categories?

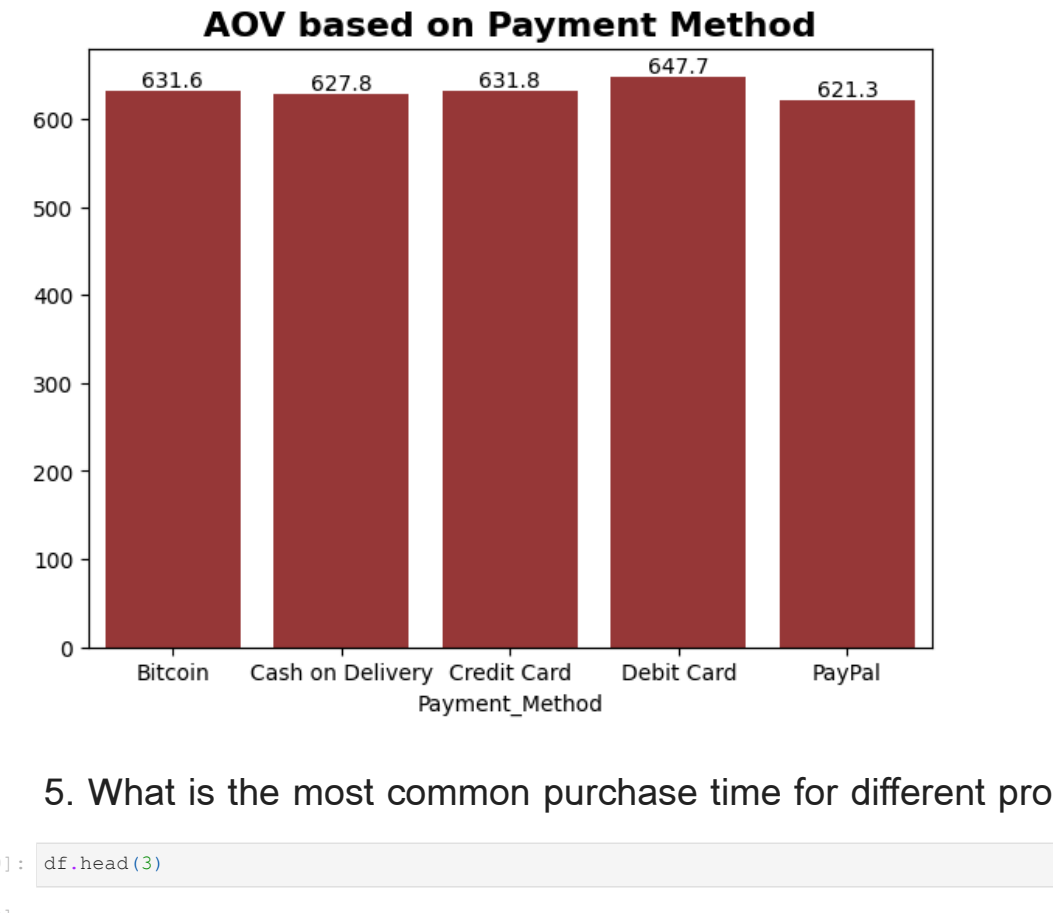
In [20]:
```python
df.head(3)
```

Out[20]:

	Order_ID	Customer_ID	Product_Category	Product_Name	Quantity	Unit_Price	Total_Price	Purchase_Date	Payment_Method	Shipping_Time_Days	Customer_Rating	Discount_Percentage	Order_Date	Revenue
0	ORD100000	CUST6193	Books	Sofa	3	97.51	292.52	2022-01-01 00:00:00	Debit Card	8	3	19	2022-01-01	292.53
1	ORD100001	CUST4745	Books	Sofa	4	265.08	1060.31	2022-01-01 01:00:00	Bitcoin	9	4	13	2022-01-01	1060.32
2	ORD100002	CUST2838	Toys	Novel	1	82.31	82.31	2022-01-01 02:00:00	Credit Card	8	5	29	2022-01-01	82.31

In [21]:
```python
df['Hour']  = df['Purchase_Date'].dt.hour
```

In [22]:
```python
mtn.figure(figsize = (8,5))
sea.boxplot(x = 'Product_Category', y = 'Hour', data = df, color = 'black')
mtn.title("Purchase Time Distribution by Product Category", fontweight = 'bold')
mtn.show()
```



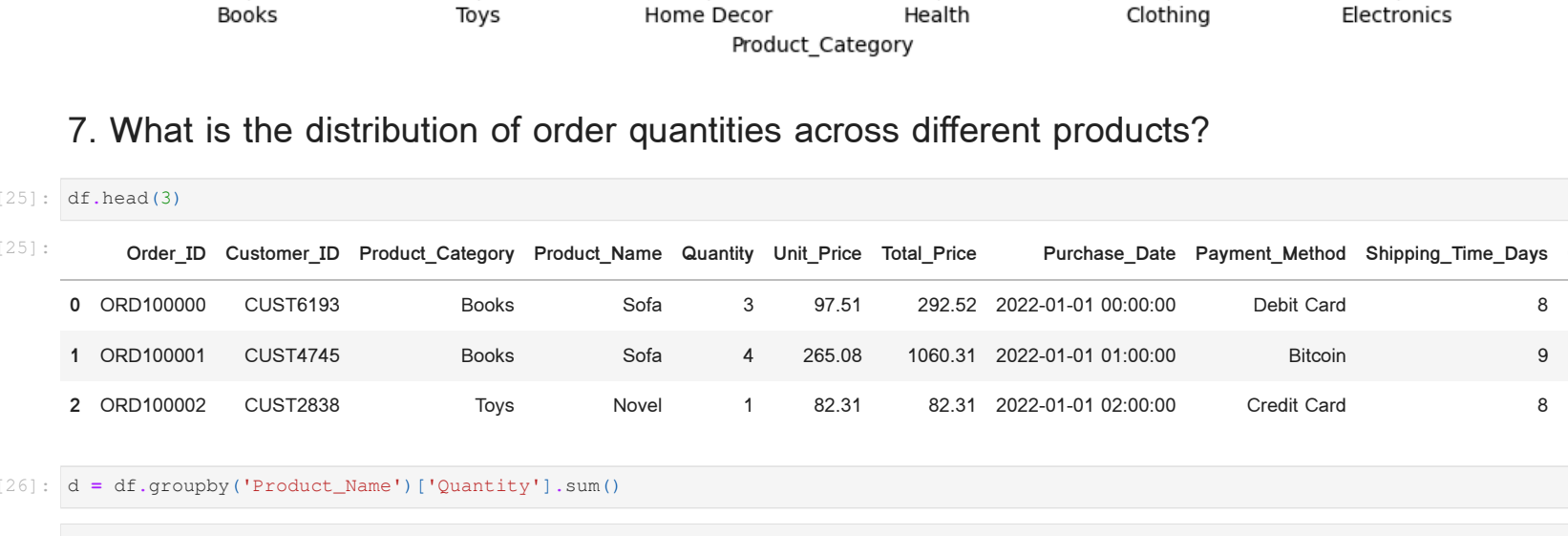
6. How do customer ratings correlate with product categories?

In [23]:
```python
df.head(3)
```

Out[23]:

	Order_ID	Customer_ID	Product_Category	Product_Name	Quantity	Unit_Price	Total_Price	Purchase_Date	Payment_Method	Shipping_Time_Days	Customer_Rating	Discount_Percentage	Order_Date	Revenue	Hour
0	ORD100000	CUST6193	Books	Sofa	3	97.51	292.52	2022-01-01 00:00:00	Debit Card	8	3	19	2022-01-01	292.53	0
1	ORD100001	CUST4745	Books	Sofa	4	265.08	1060.31	2022-01-01 01:00:00	Bitcoin	9	4	13	2022-01-01	1060.32	1
2	ORD100002	CUST2838	Toys	Novel	1	82.31	82.31	2022-01-01 02:00:00	Credit Card	8	5	29	2022-01-01	82.31	2

In [24]:
```python
mtn.figure(figsize = (12,5))
sea.boxplot(x = 'Product_Category', y = 'Customer_Rating', data = df, color = 'hotpink')
mtn.title("Customer Ratings by Product Category", fontweight = 'bold')
mtn.show()
```



7. What is the distribution of order quantities across different products?

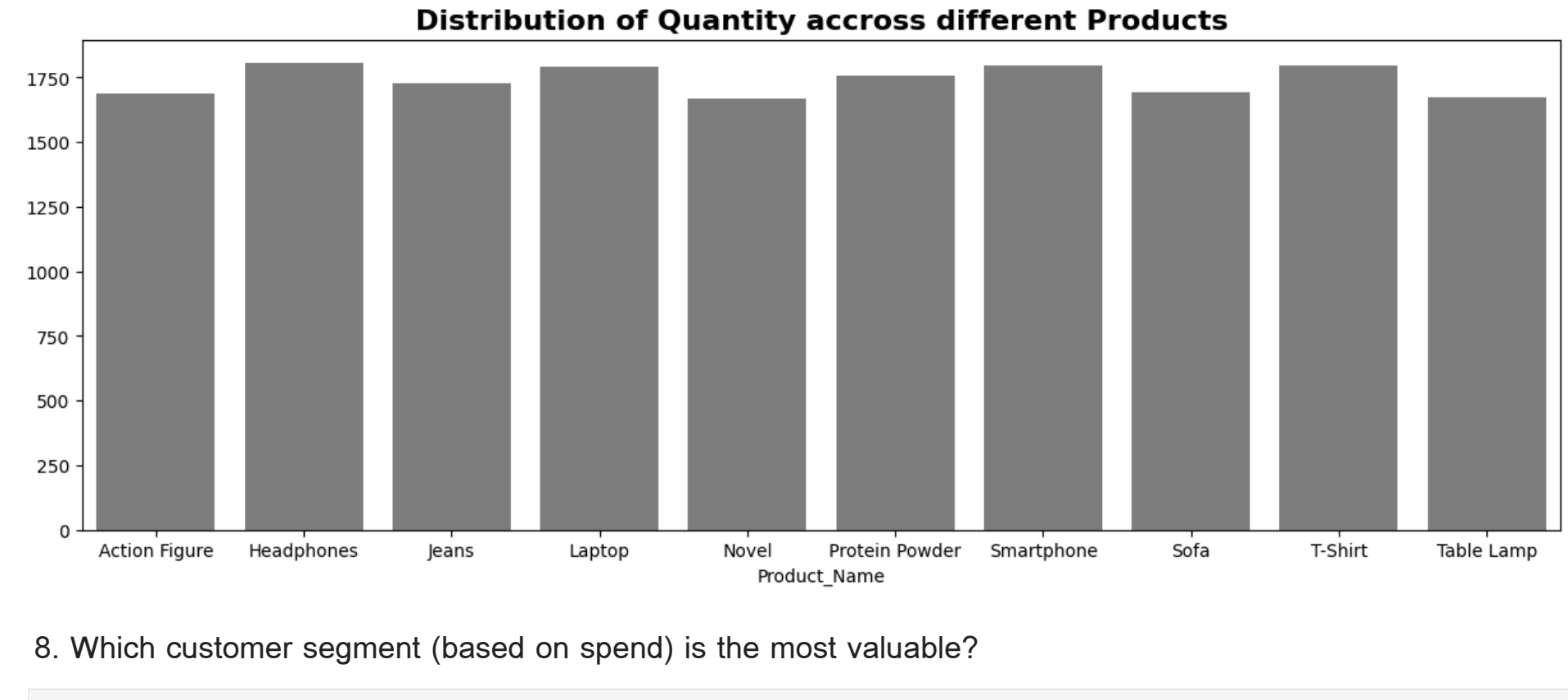
In [25]:
```python
df.head(3)
```

Out[25]:

	Order_ID	Customer_ID	Product_Category	Product_Name	Quantity	Unit_Price	Total_Price	Purchase_Date	Payment_Method	Shipping_Time_Days	Customer_Rating	Discount_Percentage	Order_Date	Revenue	Hour
0	ORD100000	CUST6193	Books	Sofa	3	97.51	292.52	2022-01-01 00:00:00	Debit Card	8	3	19	2022-01-01	292.53	0
1	ORD100001	CUST4745	Books	Sofa	4	265.08	1060.31	2022-01-01 01:00:00	Bitcoin	9	4	13	2022-01-01	1060.32	1
2	ORD100002	CUST2838	Toys	Novel	1	82.31	82.31	2022-01-01 02:00:00	Credit Card	8	5	29	2022-01-01	82.31	2

In [26]:
```python
d = df.groupby('Product_Name')['Quantity'].sum()
```

In [27]:
```python
mtn.figure(figsize =  (15,5))
sea.barplot(x = d.index, y = d.values, color = "grey")
mtn.title('Distribution of Quantity accross different Products', fontsize = 16, fontweight = 'bold')
mtn.show()
```



8. Which customer segment (based on spend) is the most valuable?

In [30]:
```python
df.head(3)
```

Out[30]:

	Order_ID	Customer_ID	Product_Category	Product_Name	Quantity	Unit_Price	Total_Price	Purchase_Date	Payment_Method	Shipping_Time_Days	Customer_Rating	Discount_Percentage	Order_Date	Revenue	Hour
0	ORD100000	CUST6193	Books	Sofa	3	97.51	292.52	2022-01-01 00:00:00	Debit Card	8	3	19	2022-01-01	292.53	0
1	ORD100001	CUST4745	Books	Sofa	4	265.08	1060.31	2022-01-01 01:00:00	Bitcoin	9	4	13	2022-01-01	1060.32	1
2	ORD100002	CUST2838	Toys	Novel	1	82.31	82.31	2022-01-01 02:00:00	Credit Card	8	5	29	2022-01-01	82.31	2

In [37]:
```python
round(df.groupby('Customer_ID')['Revenue'].sum().nlargest(7).reset_index(),1)
```

Out[37]:

	Customer_ID	Revenue
0	CUST8489	6196.3
1	CUST9033	4836.5
2	CUST3886	4586.4
3	CUST9332	4531.7
4	CUST7155	4448.6
5	CUST4283	4402.2
6	CUST2330	4276.1

9. Does a higher discount percentage lead to increased sales volume?

In [38]:
```python
df.head(2)
```

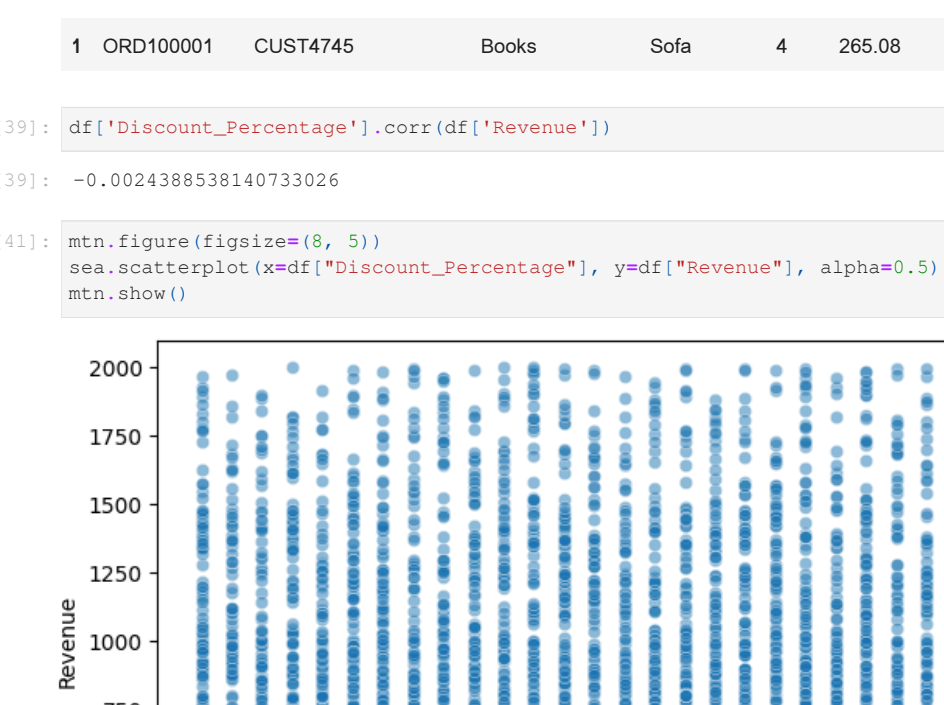
Out[38]:

	Order_ID	Customer_ID	Product_Category	Product_Name	Quantity	Unit_Price	Total_Price	Purchase_Date	Payment_Method	Shipping_Time_Days	Customer_Rating	Discount_Percentage	Order_Date	Revenue	Hour
0	ORD100000	CUST6193	Books	Sofa	3	97.51	292.52	2022-01-01 00:00:00	Debit Card	8	3	19	2022-01-01	292.53	0
1	ORD100001	CUST4745	Books	Sofa	4	265.08	1060.31	2022-01-01 01:00:00	Bitcoin	9	4	13	2022-01-01	1060.32	1

In [39]:
```python
df['Discount_Percentage'].corr(df['Revenue'])
```

Out[39]:
```
-0.0024388538140733026
```

In [41]:
```python
mtn.figure(figsize=(8, 5))
sea.scatterplot(x=df["Discount_Percentage"], y=df["Revenue"], alpha=0.5)
mtn.show()
```



There is no correlation between Revenue and Discount_Percentage, also the scatterplot seems parralel for both, so that a higher discount percentage doesn't lead to increased sales volume

10. Which payment method is associated with the highest average order value?

In [42]:
```python
df.head(3)
```

Out[42]:

	Order_ID	Customer_ID	Product_Category	Product_Name	Quantity	Unit_Price	Total_Price	Purchase_Date	Payment_Method	Shipping_Time_Days	Customer_Rating	Discount_Percentage	Order_Date	Revenue	Hour
0	ORD100000	CUST6193	Books	Sofa	3	97.51	292.52	2022-01-01 00:00:00	Debit Card	8	3	19	2022-01-01	292.53	0
1	ORD100001	CUST4745	Books	Sofa	4	265.08	1060.31	2022-01-01 01:00:00	Bitcoin	9	4	13	2022-01-01	1060.32	1
2	ORD100002	CUST2838	Toys	Novel	1	82.31	82.31	2022-01-01 02:00:00	Credit Card	8	5	29	2022-01-01	82.31	2

In [44]:
```python
round(df.groupby('Payment_Method')['Total_Price'].mean().nlargest(1).reset_index(),0)
```

Out[44]:

	Payment_Method	Total_Price
0	Debit Card	648.0

11. Identify the top 10 customers who made the highest purchases.

In [45]:
```python
df.head(2)
```

Out[45]:

	Order_ID	Customer_ID	Product_Category	Product_Name	Quantity	Unit_Price	Total_Price	Purchase_Date	Payment_Method	Shipping_Time_Days	Customer_Rating	Discount_Percentage	Order_Date	Revenue	Hour
0	ORD100000	CUST6193	Books	Sofa	3	97.51	292.52	2022-01-01 00:00:00	Debit Card	8	3	19	2022-01-01	292.53	0
1	ORD100001	CUST4745	Books	Sofa	4	265.08	1060.31	2022-01-01 01:00:00	Bitcoin	9	4	13	2022-01-01	1060.32	1

In [49]:
```python
df.groupby('Customer_ID').size().reset_index(name = 'Purchases').sort_values('Purchases', ascending = False).head(10)
```

Out[49]:

	Customer_ID	Purchases
4093	CUST8489	6
4511	CUST9260	6
1017	CUST2839	5
3912	CUST8168	5
1343	CUST3455	5
4448	CUST9147	5
2084	CUST4822	5
3350	CUST7155	5
896	CUST2623	5
603	CUST2107	5

12. What is the correlation between total price and shipping time?

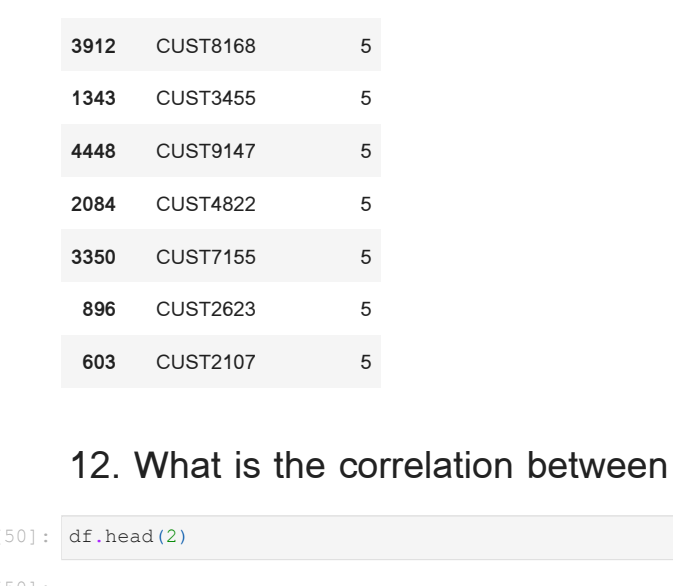
In [50]:
```python
df.head(2)
```

Out[50]:

	Order_ID	Customer_ID	Product_Category	Product_Name	Quantity	Unit_Price	Total_Price	Purchase_Date	Payment_Method	Shipping_Time_Days	Customer_Rating	Discount_Percentage	Order_Date	Revenue	Hour
0	ORD100000	CUST6193	Books	Sofa	3	97.51	292.52	2022-01-01 00:00:00	Debit Card	8	3	19	2022-01-01	292.53	0
1	ORD100001	CUST4745	Books	Sofa	4	265.08	1060.31	2022-01-01 01:00:00	Bitcoin	9	4	13	2022-01-01	1060.32	1

In [62]:
```python
correlation = df[['Shipping_Time_Days', 'Total_Price']].corr()
```

In [63]:
```python
mtn.figure(figsize = (6,5))
sea.heatmap(correlation, annot = True, cmap = 'coolwarm')
mtn.show()
```



There is no correlation between Shipping time and Total Price

13. Which product categories see the highest price fluctuations?

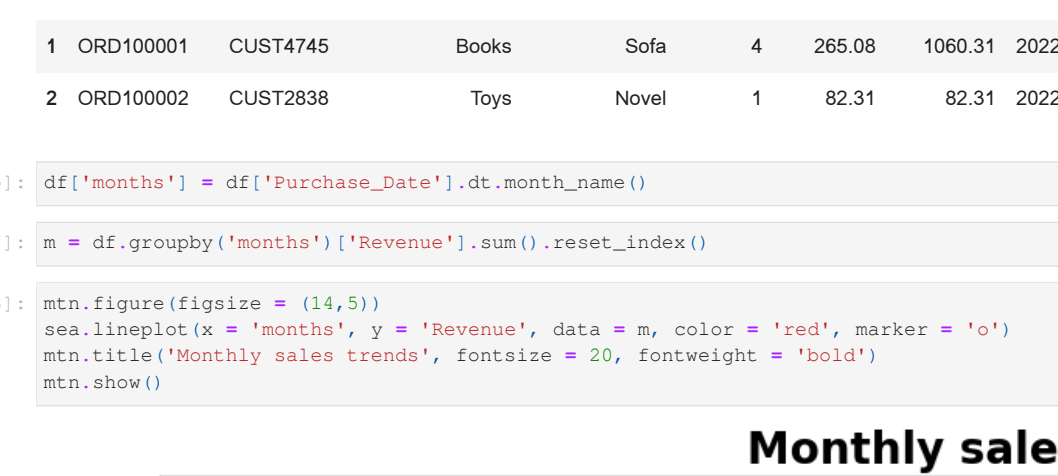
In [64]:
```python
df.head(2)
```

Out[64]:

	Order_ID	Customer_ID	Product_Category	Product_Name	Quantity	Unit_Price	Total_Price	Purchase_Date	Payment_Method	Shipping_Time_Days	Customer_Rating	Discount_Percentage	Order_Date	Revenue	Hour
0	ORD100000	CUST6193	Books	Sofa	3	97.51	292.52	2022-01-01 00:00:00	Debit Card	8	3	19	2022-01-01	292.53	0
1	ORD100001	CUST4745	Books	Sofa	4	265.08	1060.31	2022-01-01 01:00:00	Bitcoin	9	4	13	2022-01-01	1060.32	1

In [69]:
```python
y = round(df.groupby('Product_Category')['Unit_Price'].std().sort_values(ascending = False).reset_index(),1)
```

In [72]:
```python
mtn.figure(figsize = (8,4))
ax = sea.barplot(x = 'Product_Category', y = 'Unit_Price', data = y, color = 'brown')
ax.bar_label(ax.containers[0])
mtn.title('Highest Price Fluctuations')
mtn.show()
```



14. How does seasonality impact sales? (e.g., monthly sales trends)

In [73]:
```python
df.head(3)
```

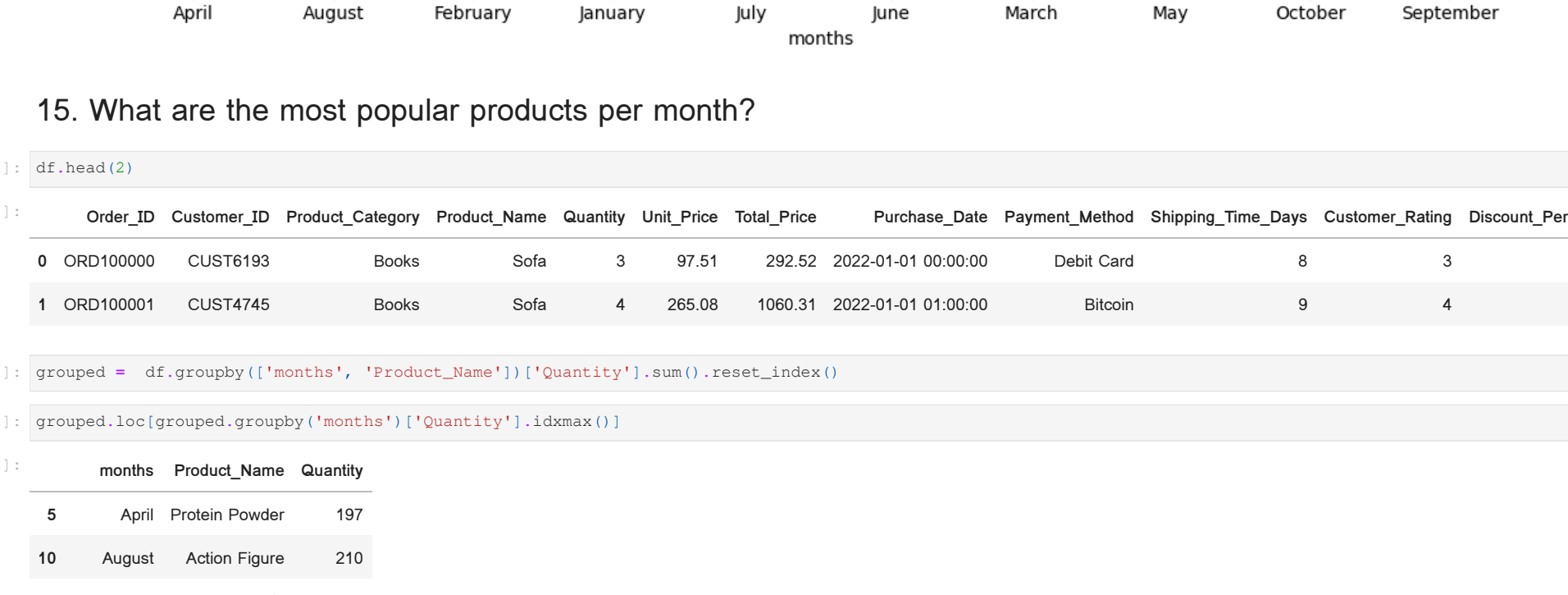
Out[73]:

	Order_ID	Customer_ID	Product_Category	Product_Name	Quantity	Unit_Price	Total_Price	Purchase_Date	Payment_Method	Shipping_Time_Days	Customer_Rating	Discount_Percentage	Order_Date	Revenue	Hour
0	ORD100000	CUST6193	Books	Sofa	3	97.51	292.52	2022-01-01 00:00:00	Debit Card	8	3	19	2022-01-01	292.53	0
1	ORD100001	CUST4745	Books	Sofa	4	265.08	1060.31	2022-01-01 01:00:00	Bitcoin	9	4	13	2022-01-01	1060.32	1
2	ORD100002	CUST2838	Toys	Novel	1	82.31	82.31	2022-01-01 02:00:00	Credit Card	8	5	29	2022-01-01	82.31	2

In [75]:
```python
df['months'] = df['Purchase_Date'].dt.month_name()
```

In [77]:
```python
m = df.groupby('months')['Revenue'].sum().reset_index()
```

In [83]:
```python
mtn.figure(figsize = (14,5))
sea.lineplot(x = 'months', y = 'Revenue', data = m, color = 'red', marker = 'o')
mtn.title('Monthly sales trends', fontsize = 20, fontweight = 'bold')
mtn.show()
```



15. What are the most popular products per month?

In [84]:
```python
df.head(2)
```

Out[84]:

	Order_ID	Customer_ID	Product_Category	Product_Name	Quantity	Unit_Price	Total_Price	Purchase_Date	Payment_Method	Shipping_Time_Days	Customer_Rating	Discount_Percentage	Order_Date	Revenue	Hour	months
0	ORD100000	CUST6193	Books	Sofa	3	97.51	292.52	2022-01-01 00:00:00	Debit Card	8	3	19	2022-01-01	292.53	0	January
1	ORD100001	CUST4745	Books	Sofa	4	265.08	1060.31	2022-01-01 01:00:00	Bitcoin	9	4	13	2022-01-01	1060.32	1	January

In [91]:
```python
grouped =  df.groupby(['months', 'Product_Name'])['Quantity'].sum().reset_index()
```

In [92]:
```python
grouped.loc[grouped.groupby('months')['Quantity'].idxmax()]
```

Out[92]:

	months	Product_Name	Quantity
5	April	Protein Powder	197
10	August	Action Figure	210
27	February	Sofa	199
37	January	Sofa	220
46	July	Smartphone	206
56	June	Smartphone	233
68	March	T-Shirt	215
76	May	Smartphone	208
82	October	Jeans	145
91	September	Headphones	207

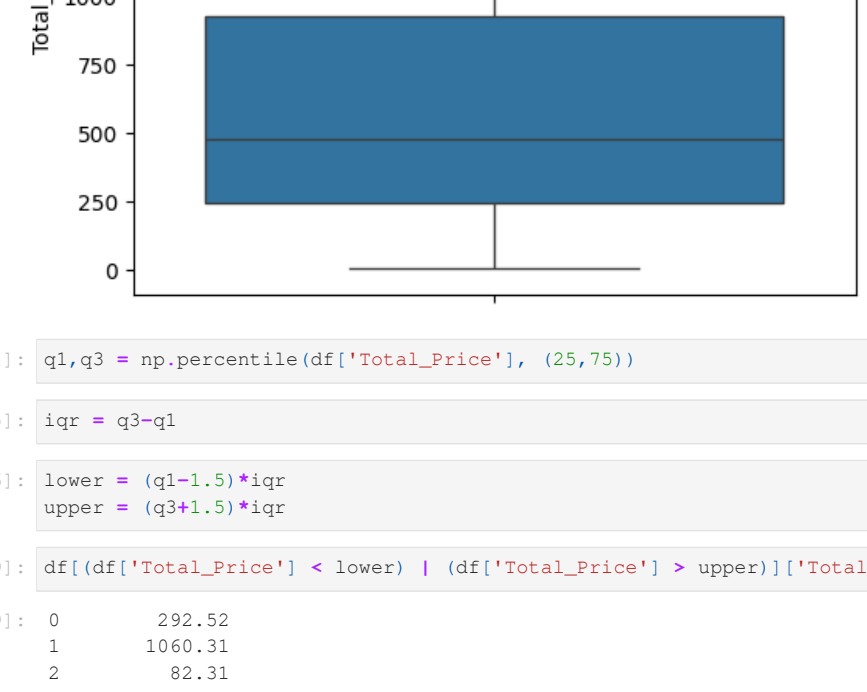
16. Can we detect any outliers in pricing or order values?

In [89]:
```python
df.head(2)
```

Out[89]:

	Order_ID	Customer_ID	Product_Category	Product_Name	Quantity	Unit_Price	Total_Price	Purchase_Date	Payment_Method	Shipping_Time_Days	Customer_Rating	Discount_Percentage	Order_Date	Revenue	Hour	months
0	ORD100000	CUST6193	Books	Sofa	3	97.51	292.52	2022-01-01 00:00:00	Debit Card	8	3	19	2022-01-01	292.53	0	January
1	ORD100001	CUST4745	Books	Sofa	4	265.08	1060.31	2022-01-01 01:00:00	Bitcoin	9	4	13	2022-01-01	1060.32	1	January

In [93]:
```python
mtn.figure(figsize = (6,5))
sea.boxplot(df['Total_Price'])
mtn.title('Outliers in Price')
mtn.show()
```



In [94]:
```python
q1,q3 = np.percentile(df['Total_Price'], (25,75))
```

In [95]:
```python
iqr = q3-q1
```

In [96]:
```python
lower = (q1-1.5)*iqr
upper = (q3+1.5)*iqr
```

In [99]:
```python
df[(df['Total_Price'] < lower) | (df['Total_Price'] > upper)]['Total_Price']
```

Out[99]:
```
0        292.52
1       1060.31
2         82.31
3       1285.97
4        564.72
         ...   
6995     688.90
6996    1122.04
6997    1156.56
6998     105.59
6999     813.18
Name: Total_Price, Length: 7000, dtype: float64
```

17. What is the impact of discounts on customer ratings?

In [100…
```python
df.head(2)
```

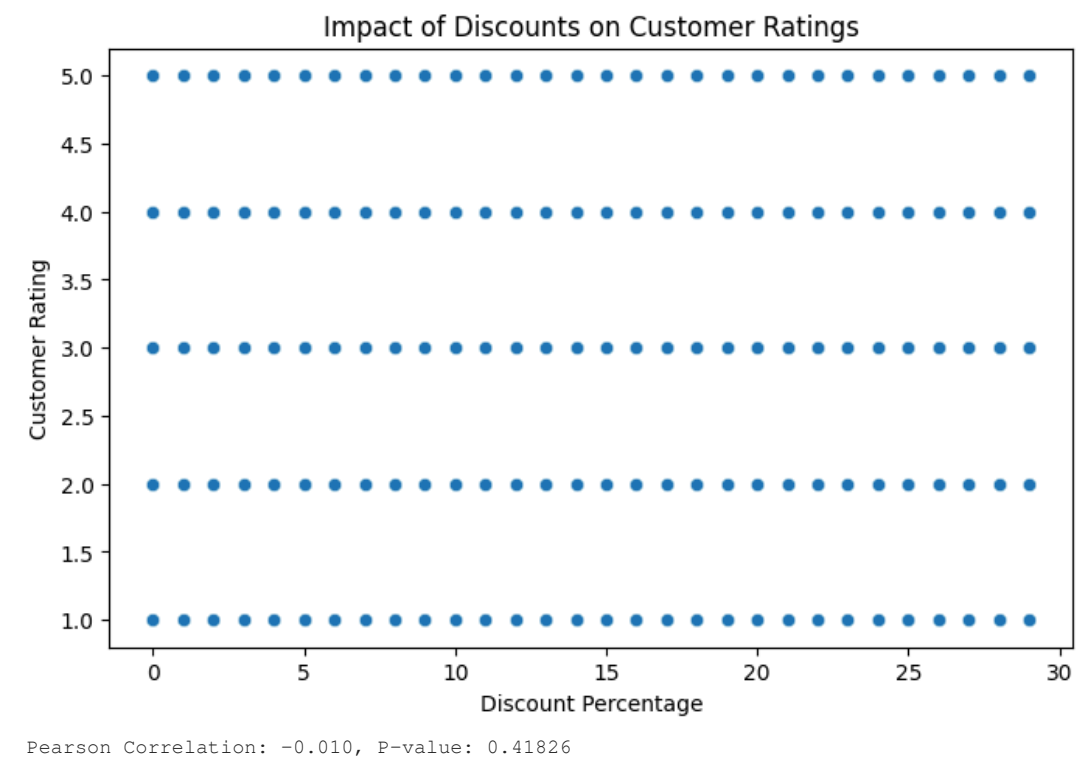
Out[100…

	Order_ID	Customer_ID	Product_Category	Product_Name	Quantity	Unit_Price	Total_Price	Purchase_Date	Payment_Method	Shipping_Time_Days	Customer_Rating	Discount_Percentage	Order_Date	Revenue	Hour	months
0	ORD100000	CUST6193	Books	Sofa	3	97.51	292.52	2022-01-01 00:00:00	Debit Card	8	3	19	2022-01-01	292.53	0	January
1	ORD100001	CUST4745	Books	Sofa	4	265.08	1060.31	2022-01-01 01:00:00	Bitcoin	9	4	13	2022-01-01	1060.32	1	January

In [101…
```python
from scipy.stats import pearsonr

mtn.figure(figsize=(8, 5))
sea.scatterplot(x=df["Discount_Percentage"], y=df["Customer_Rating"], alpha=0.5)
mtn.title("Impact of Discounts on Customer Ratings")
mtn.xlabel("Discount Percentage")
mtn.ylabel("Customer Rating")
mtn.show()

# Compute Pearson correlation
correlation, p_value = pearsonr(df["Discount_Percentage"], df["Customer_Rating"])
print(f"Pearson Correlation: {correlation:.3f}, P-value: {p_value:.5f}")
```



```
Pearson Correlation: -0.010, P-value: 0.41826
```

Discounts and customer ratings are not related in this dataset.

Even if a company gives higher discounts, it doesn’t strongly impact customer ratings (either positively or negatively).

Any small negative correlation (-0.010) is insignificant and likely due to randomness.

In [ ]: